Ali Saeed Chaudhary

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Linkedin | Portfolio

EDUCATION

Iowa State University Ames, Iowa
B.S., Mechanical Engineering - Cyber-Physical Systems Minor. ***GPA: 4.0.*** *May 2028*
Korea University Seoul, South Korea
Study Abroad Program. *June 2026*
Studied fundamentals of robotics and mechatronics, developed reinforcement learning algorithms using PPO, TRPO, and VLA Models for robotics applications, Gained coursework knowledge in nuclear power and nuclear energy

EXPERIENCE

Research Assistant Aug 2026 – Present
Iowa State University, Coordinated Systems Lab *Ames, Iowa*
- Partnered with RTX Corporation to develop an air traffic prediction network
- Researching airspace trajectory prediction with cascading confidence intervals using RNNs
- Writing code for neural network models and conducting research into available training data; building datasets and collecting relevant literature in the project's preliminary phase

Radiology IT Support Coordinator Nov 2024 – Feb 2026
Concord Medical Management, LLC *Flower Mound, Texas (remote)*
- Coordinated live data flow between remote health facilities and radiologists, resolving technical issues under tight deadlines
- Optimized system escalation workflows and communications to ensure timely medical IT support

ACTIVITIES & LEADERSHIP

Cardinal Space Mining, NASA Lunabotics Aug 2026 - Present
Collections Sub-Team Member *Ames, Iowa*
- Leading dust-cover design for the collections sub-system of a lunar regolith mining rover
- Improving robustness over the prior year's design while maintaining functionality and manufacturability

Cyclone Rocketry, sponsored by Boeing Oct 2024 - Dec 2025
Mechanical Sub-Team Member *Ames, Iowa*
- proposed and lead the designing of an airbrakes flight mechanics system
- modeled components in SolidWorks and ran FEA simulations in Ansys

PROJECTS

RL-Trained Robotic Hand May – June 2026
- Trained a reinforcement learning policy in Isaac Sim to control a robotic hand using reference data from the HO-Cap (NeurIPS 2025) dataset
- Designed the reward function, weighting fingertip and palm contact points to encourage stable grip formation and penalizing unnecessary motion
- Configured observations including robot and reference hand keypoints relative to the object, joint positions, and fingertip contact forces

Substrate Rubbing Machine May – August 2024
- Led design, construction, and testing of a custom substrate rubbing machine for the FMOD Research Lab, automating a previously manual process
- Designed the machine in Onshape, designed a custom PCB for the button panel, and 3D-printed control surfaces
- Implemented an automated rubbing routine with manual override controls; deployed to the lab after testing

Private Home Server June 2026 – Present
- Building a private media server for storing, hosting, and distributing services and files
- Configured remote access through an Oracle VPS using Pangolin and Traefik as a reverse-proxy, routing multiple services behind a single-entry point

SKILLS

CAD & Simulation: SolidWorks, Onshape, Fusion360, Isaac Sim, Ansys FEA, CAM
Programming & ML: Python, RNNs, PPO, SAC, VLA Models, Robotics, ROS
Systems & Tools: Debian Linux, TailScale, Network Systems Management
Manufacturing: CNC Milling, Welding (TIG, MIG), Metal Working, 3D-Printing